Notes: ***Create a Jira ticket for this assignment, work on the tasks listed below, and submit the task completion document by EOD with clear steps and required evidence (outputs/screenshots)***

1. Install nginx and run nginx on port number 81.
2. Deploy a sample index.html file on nginx.
3. Install Apache and run Apache on port number 82
4. Deploy a sample index.html file on Apache.
5. Install Apache tomcat on port number 8082
6. Deploy a sample app on webapps
7. Create a tomcat.service file for tomcat.
8. Configure HA Proxy server

1 **Install Nginx and run Nginx on port number 81**

- **sudo nano /etc/nginx/sites-available/default**
    - **Use:** Opens the Nginx default site configuration file to modify the listening port to port 81.
- **sudo nginx -t && sudo systemctl restart nginx**
    - **Use:** Tests the Nginx configuration syntax for correctness and restarts the Nginx service to apply port changes.

```
Unit apach.service could not be found.
vboxuser@Ubuntu-Lab:~/Desktop$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabl>
     Active: active (running) since Wed 2026-10-14 10:55:50 UTC; 25s ago
 Invocation: 33592706ee6243cfb11a7272b9adca2f
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 4980 (apache2)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes serv>
      Tasks: 55 (limit: 4024)
     Memory: 5.3M (peak: 5.5M)
        CPU: 89ms
     CGroup: /system.slice/apache2.service
             ├─4980 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─4983 /usr/sbin/apache2 -k start -DFOREGROUND
             └─4984 /usr/sbin/apache2 -k start -DFOREGROUND

Oct 14 10:55:50 Ubuntu-Lab systemd[1]: Starting apache2.service - The Apache HTTP Se>
Oct 14 10:55:50 Ubuntu-Lab apachectl[4980]: AH00558: apache2: Could not reliably det>
Oct 14 10:55:50 Ubuntu-Lab systemd[1]: Started apache2.service - The Apache HTTP Ser>

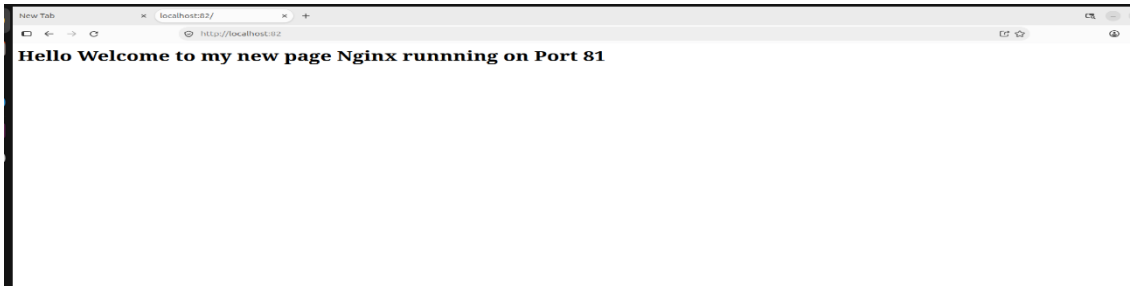
vboxuser@Ubuntu-Lab:~/Desktop$ curl http://localhost:82
<h1>Hello Welcome to my new page Nginx runnning on Port 81</h1>
vboxuser@Ubuntu-Lab:~/Desktop$
```

```
sudo: 'nani': command not found
vboxuser@Ubuntu-Lab:~/Desktop$ sudo nano /etc/nginx/sites-available/default
vboxuser@Ubuntu-Lab:~/Desktop$ sudo bash -c 'echo "<h1>Hello Welcome to my new page Ngi
runnning on Port 81</h1>" > /var/www/html/index.html'
vboxuser@Ubuntu-Lab:~/Desktop$ sudo nginx -t && sudo systemctl restart nginx
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
Job for nginx.service failed because the control process exited with error code.
See "systemctl status nginx.service" and "journalctl -xeu nginx.service" for details.
vboxuser@Ubuntu-Lab:~/Desktop$ sudo ss -tuln | grep ':80'
tcp    LISTEN 0      100                *:8080             *:*
tcp    LISTEN 0      1      [::ffff:127.0.0.1]:8005        *:*
tcp    LISTEN 0      511                *:80               *:*
vboxuser@Ubuntu-Lab:~/Desktop$ sudo system stop apache2
sudo: 'system': command not found
vboxuser@Ubuntu-Lab:~/Desktop$ sudo systemctl stop apache2
vboxuser@Ubuntu-Lab:~/Desktop$ sudo ss -tuln | grep ':80'
tcp    LISTEN 0      100                *:8080             *:*
tcp    LISTEN 0      1      [::ffff:127.0.0.1]:8005        *:*
vboxuser@Ubuntu-Lab:~/Desktop$ sudo systemctl restart nginx
vboxuser@Ubuntu-Lab:~/Desktop$ sudo nano /etc/apache2/ports.conf
vboxuser@Ubuntu-Lab:~/Desktop$ sudo nano /etc/apache2/sites-available/000-default.conf
vboxuser@Ubuntu-Lab:~/Desktop$ sudo systemctl restart apache2
vboxuser@Ubuntu-Lab:~/Desktop$ sudo systemctl status apach
```

Deploy a sample index.html file on Nginx

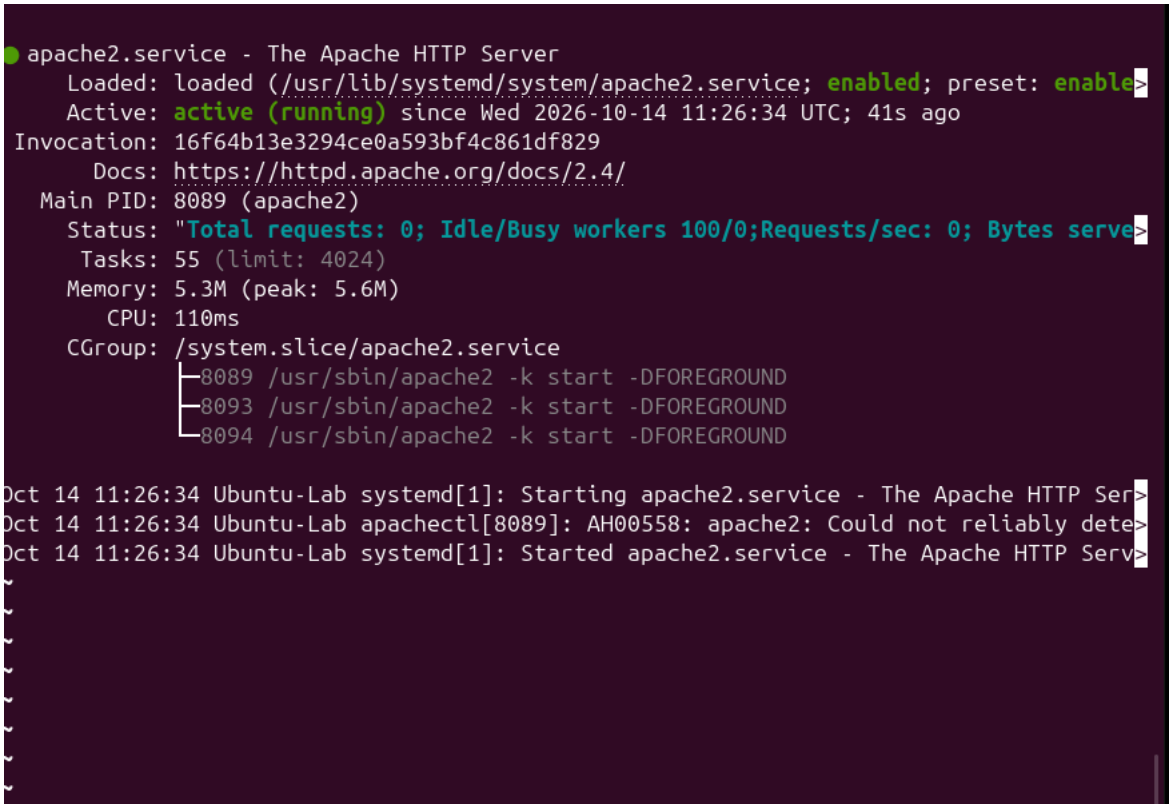
- **sudo bash -c 'echo "..." > /var/www/html/index.html'**

- **Use:** Creates and deploys a sample HTML page directly into the default Nginx web root directory (/var/www/html/).



Install Apache and run Apache on port number 82

- **sudo systemctl stop apache2**
  - **Use:** Stops the Apache2 service to resolve port conflicts or manage active processes.
- **sudo nano /etc/apache2/ports.conf**
  - **Use:** Edits Apache's global port configuration file to change the listening port.
- **sudo nano /etc/apache2/sites-available/000-default.conf**
  - **Use:** Modifies the default Apache virtual host configuration file to bind Apache to port 82.
- **sudo systemctl restart apache2**
  - **Use:** Restarts the Apache service to apply the new port 82 configuration.
- **sudo systemctl status apache2**
  - **Use:** Verifies that the Apache service is actively running.



4 **Install Apache and run Apache on port number 82**

- **sudo systemctl stop apache2**
  - **Use:** Stops the Apache2 service to resolve port conflicts or manage active processes.
- **sudo nano /etc/apache2/ports.conf**
  - **Use:** Edits Apache's global port configuration file to change the listening port.
- **sudo nano /etc/apache2/sites-available/000-default.conf**
  - **Use:** Modifies the default Apache virtual host configuration file to bind Apache to port 82.
- **sudo systemctl restart apache2**
  - **Use:** Restarts the Apache service to apply the new port 82 configuration.
- **sudo systemctl status apache2**
  - **Use:** Verifies that the Apache service is actively running.




```
vboxuser@Ubuntu-Lab:~/Desktop$ sudo bash -c 'echo "<h1>Hello from Apache HTTP Server r
unning on port 82 </h1>" > /var/www/html/index.html'
vboxuser@Ubuntu-Lab:~/Desktop$ cat /var/www/html/index.html
<h1>Hello from Apache HTTP Server running on port 82 </h1>
vboxuser@Ubuntu-Lab:~/Desktop$ curl http://localhost:82
<h1>Hello from Apache HTTP Server running on port 82 </h1>
vboxuser@Ubuntu-Lab:~/Desktop$
```

5 **Install Apache Tomcat on port number 8082**

- **sudo nano /opt/tomcat/conf/server.xml**
    - **Use:** Opens the Tomcat server configuration file to verify or configure the port setting.
- **sudo grep 'port="8082"' /opt/tomcat/conf/server.xml**
    - **Use:** Searches the XML configuration file to confirm that the HTTP connector is successfully set to port 8082.

```
drwx------ 3 root     root           60 Oct 14 10:24 systemd-private-158cebcd7574489b8
dbd93b4b1b1cd3e-switcheroo-control.service-dqrgWt
drwx------ 3 root     root           60 Oct 14 10:24 systemd-private-158cebcd7574489b8
dbd93b4b1b1cd3e-systemd-logind.service-3kiR7a
drwx------ 3 root     root           60 Oct 14 10:25 systemd-private-158cebcd7574489b8
dbd93b4b1b1cd3e-upower.service-qGTAtK
vboxuser@Ubuntu-Lab:/tmp$ sudo nano /opt/tomcat/conf/server.xml
vboxuser@Ubuntu-Lab:/tmp$ grep 'port="8082"' /opt/tomcat/conf/server.xml
grep: /opt/tomcat/conf/server.xml: Permission denied
vboxuser@Ubuntu-Lab:/tmp$ sudo grep 'port="8082"' /opt/tomcat/conf/server.xml
    <Connector port="8082" protocol="HTTP/1.1"
vboxuser@Ubuntu-Lab:/tmp$
```

6 **Deploy a sample app on webapps**

- **sudo bash -c 'echo "..." > /opt/tomcat/apache-tomcat-11.0.0-M20/webapps/sampleapp/index.html'**
    - **Use:** Creates a sample application folder (sampleapp) and deploys an index.html file inside Tomcat's webapps directory.
- **sudo chown -R tomcat:tomcat /opt/tomcat**
    - **Use:** Recursively changes the file ownership of the Tomcat directory to the tomcat user and group so the service has read/write access.
- **ls -l /opt/tomcat/.../index.html**

- **Use:** Checks file permissions and details to verify the application deployment.

```
vboxuser@Ubuntu-Lab:/tmp$ sudo bash -c 'echo "<h1>hello from apache tomact 11 sample
app on port 8082</h1>" > /opt/tomcat/apache-tomcat-11.0.0-M20/webapps/sampleapp/index
.html'
vboxuser@Ubuntu-Lab:/tmp$ sudo cat /opt/tomcat/apache-tomcat-11.0.0-M20/webapps/sampl
eapp/index.html
<h1>hello from apache tomact 11 sample app on port 8082</h1>
vboxuser@Ubuntu-Lab:/tmp$ sudo chown -R tomcat:tomcat /opt/tomcat
vboxuser@Ubuntu-Lab:/tmp$ ls -l /opt/tomcat/apache-tomcat-11.0.0-M20/webapps/sampleap
p/index.html
-rw-r--r-- 1 tomcat tomcat 61 Oct 14 13:53 /opt/tomcat/apache-tomcat-11.0.0-M20/webap
ps/sampleapp/index.html
vboxuser@Ubuntu-Lab:/tmp$
```

7 **Create a tomcat.service file for Tomcat**

- **sudo nano /etc/systemd/system/tomcat.service**
  - **Use:** Creates and edits a custom systemd service configuration file to manage Tomcat as a system service.
- **sudo systemctl daemon-reload**
  - **Use:** Refreshes the systemd manager to recognize the newly created tomcat.service file.
- **sudo systemctl start tomcat**
  - **Use:** Starts the Apache Tomcat service using the systemd manager.

```
vboxuser@Ubuntu-Lab:~/Desktop$ sudo nano /etc/systemd/system/tomcat.service
vboxuser@Ubuntu-Lab:~/Desktop$ sudo cat  /etc/systemd/system/tomcat.service
[Unit]
Description=Apache Tomcat 11 Web Application Container
After=network.target

[Service]
Type=forking

User=tomcat
Group=tomcat

Environment="JAVA_HOME=/usr/lib/jvm/default-java"
Environment="CATALINA_PID=/opt/tomcat/apache-tomcat-11.0.0-M20/temp/tomcat.pid"
Environment="CATALINA_HOME=/opt/tomcat/apache-tomcat-11.0.0-M20"
Environment="CATALINA_BASE=/opt/tomcat/apache-tomcat-11.0.0-M20"
Environment="CATALINA_OPTS=-Xms512M -Xmx1024M -server -XX:+UseParallelGC"

ExecStart=/opt/tomcat/apache-tomcat-11.0.0-M20/bin/startup.sh
ExecStop=/opt/tomcat/apache-tomcat-11.0.0-M20/bin/shutdown.sh

RestartSec=10
Restart=always

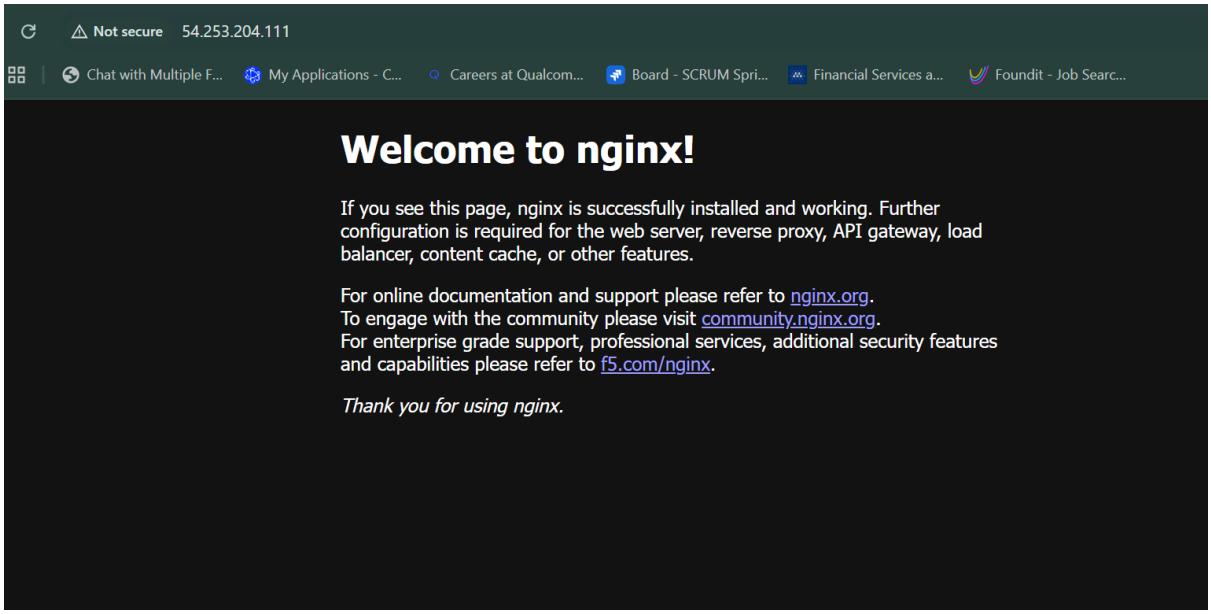
[Install]
WantedBy=multi-user.target
vboxuser@Ubuntu-Lab:~/Desktop$ sudo systemctl daemon-reload
vboxuser@Ubuntu-Lab:~/Desktop$ sudo systemctl start tomcat
```

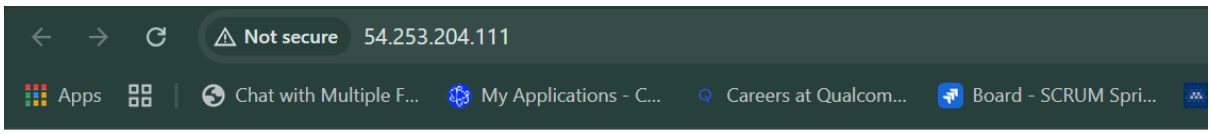
8 Ha Proxy

Configure HA Proxy server

(Note: The document screenshots for task 8 display browser verification outputs showing Nginx and load-balanced/proxy responses on public IP 54.253.204.111, but specific configuration file commands for HAProxy are part of the broader verification workflow steps.)

- **sudo ss -tulpn | grep ':80'** (or similar port-checking commands)
    - **Use:** Inspects active network sockets and listening ports to troubleshoot and verify proxy/web server port bindings.


Not secure 54.253.204.111
Chat with Multiple F...
My Applications - C...
Careers at Qualcom...
Board - SCRUM Spri...
Financial Services a...
Foundit - Job Searc...
Welcome to nginx!
If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.
For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.
Thank you for using nginx.


Not secure 54.253.204.111
Apps
Chat with Multiple F...
My Applications - C...
Careers at Qualcom...
Board - SCRUM Spri...

It works!